

Numerical Analysis – Lecture 9

6.6 The Newton interpolation formula

Recalling that $f[x_i] = f(x_i)$, the recursive formula allows for fast evaluation of the *divided difference table*

$$\begin{array}{ccccccc} f[x_0] & \rightarrow & f[x_0, x_1] & \rightarrow & f[x_0, x_1, x_2] & \rightarrow & f[x_0, x_1, x_2, x_3] & \rightarrow & \cdots \\ & \nearrow & & \nearrow & & \nearrow & & & \\ f[x_1] & \rightarrow & f[x_1, x_2] & \rightarrow & f[x_1, x_2, x_3] & \rightarrow & \cdots & & \\ & \vdots & & & & & & & \\ f[x_n] & & & & & & & & \end{array}$$

This can be done in $\mathcal{O}(n^2)$ operations.

We now provide an alternative representation of the interpolating polynomial.. Again, $f(x_i)$, $i = 0, 1, \dots, k$, are given and we seek $p \in \mathbb{P}_k[x]$ such that $p(x_i) = f(x_i)$, $i = 0, \dots, k$.

Theorem Suppose that $x_0, x_1, \dots, x_k$ are pairwise distinct. The polynomial

$$p_k(x) := f[x_0] + f[x_0, x_1](x - x_0) + \cdots + f[x_0, x_1, \dots, x_k] \prod_{i=0}^{k-1} (x - x_i) \in \mathbb{P}_k[x]$$

obeys $p_k(x_i) = f(x_i)$, $i = 0, 1, \dots, k$.

Proof. By induction on k . The statement is obvious for $k = 0$ and we suppose that it is true for k . We now prove that $p_{k+1}(x) - p_k(x) = f[x_0, x_1, \dots, x_{k+1}] \prod_{i=0}^k (x - x_i)$. Clearly, $p_{k+1} - p_k \in \mathbb{P}_{k+1}[x]$ and the coefficient of x^{k+1} therein is, by definition, $f[x_0, \dots, x_{k+1}]$. Moreover, $p_{k+1}(x_i) - p_k(x_i) = 0$, $i = 0, 1, \dots, k$, hence it is a multiple of $\prod_{i=0}^k (x - x_i)$, and this proves the asserted form of $p_{k+1} - p_k$. The explicit form of p_{k+1} follows. $\square$

We obtain the *Newton interpolation formula*, which requires only the top row of the divided difference table. It has several advantages over Lagrange's. In particular, its evaluation at a given point x (provided that divided differences are known) requires just $\mathcal{O}(k)$ operations, as long as we do it by the *Horner scheme*

$$\begin{aligned} p_k(x) &= \{ \{ \{ f[x_0, \dots, x_k](x - x_{k-1}) + f[x_1, \dots, x_{k-1}] \} \times (x - x_{k-2}) + f[x_0, \dots, x_{k-2}] \} \\ &\quad \times (x - x_3) + \cdots \} + f[x_0]. \end{aligned}$$

7 Orthogonal polynomials

7.1 Orthogonality amongst functions

We have already seen the scalar product $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i y_i$, acting on $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Likewise, given $w_1, w_2, \dots, w_n > 0$, we may define $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n w_i x_i y_i$. In general, a *scalar* (or *inner*) *product* is any function $\mathbb{V} \times \mathbb{V} \rightarrow \mathbb{R}$, where $\mathbb{V}$ is a vector space over the reals, subject to the following three axioms: **symmetry**: $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle \forall \mathbf{x}, \mathbf{y} \in \mathbb{V}$; **nonnegativity**: $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0 \forall \mathbf{x} \in \mathbb{V}$ and $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ iff $\mathbf{x} = \mathbf{0}$; and **linearity**: $\langle a\mathbf{x} + b\mathbf{y}, \mathbf{z} \rangle = a\langle \mathbf{x}, \mathbf{z} \rangle + b\langle \mathbf{y}, \mathbf{z} \rangle \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{V}, a, b \in \mathbb{R}$. Any scalar product defines *orthogonality*: $\mathbf{x}, \mathbf{y} \in \mathbb{V}$ are orthogonal if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$.

Let $\mathbb{V} = C[a, b]$, $w \in \mathbb{V}$ be a fixed *positive* function and define $\langle f, g \rangle := \int_a^b w(x)f(x)g(x) dx$ for all $f, g \in \mathbb{V}$. It is easy to verify all three axioms of the scalar product.

7.2 Orthogonal polynomials – definition, existence, uniqueness

We say that $p_n \in \mathbb{P}_n[x]$ is the *n*th *orthogonal polynomial* if $\langle p_n, p \rangle = 0$ for all $p \in \mathbb{P}_{n-1}[x]$. [Note: different inner products lead to different orthogonal polynomials.] A polynomial in $\mathbb{P}_n[x]$ is *monic* if the coefficient of x^n therein is one.

Theorem For every $n \in \mathbb{Z}^+$ there exists a unique monic orthogonal polynomial of degree n . Moreover, any $p \in \mathbb{P}_n[x]$ can be expanded as a linear combination of $p_0, p_1, \dots, p_n$,

Proof. We let $p_0(x) \equiv 1$ and prove the theorem by induction on n . Thus, suppose that $p_0, p_1, \dots, p_n$ have been already derived consistently with both assertions of the theorem and let $q(x) := x^{n+1} \in \mathbb{P}_{n+1}[x]$. Guided by the *Gram-Schmidt algorithm*, we choose

$$p_{n+1}(x) = q(x) - \sum_{k=0}^n \frac{\langle q, p_k \rangle}{\langle p_k, p_k \rangle} p_k(x), \quad x \in \mathbb{R}. \quad (7.1)$$

Clearly, $p_{n+1} \in \mathbb{P}_{n+1}[x]$ and it is monic (since all the terms in the sum are of lower degree). Let $m \in \{0, 1, \dots, n\}$. It follows from (7.1) and the induction hypothesis that

$$\langle p_{n+1}, p_m \rangle = \langle q, p_m \rangle - \sum_{k=0}^n \frac{\langle q, p_k \rangle}{\langle p_k, p_k \rangle} \langle p_k, p_m \rangle = \langle q, p_m \rangle - \frac{\langle q, p_m \rangle}{\langle p_m, p_m \rangle} \langle p_m, p_m \rangle = 0.$$

Hence, p_{n+1} is orthogonal to $p_0, \dots, p_n$. Consequently, according to the second inductive assertion, it is orthogonal to all $p \in \mathbb{P}_n[x]$.

To prove uniqueness, we suppose the existence of two monic orthogonal polynomials $p_{n+1}, \tilde{p}_{n+1} \in \mathbb{P}_{n+1}[x]$. Let $p := p_{n+1} - \tilde{p}_{n+1} \in \mathbb{P}_n[x]$, hence $\langle p_{n+1}, p \rangle = \langle \tilde{p}_{n+1}, p \rangle = 0$, and this implies

$$0 = \langle p_{n+1}, p \rangle - \langle \tilde{p}_{n+1}, p \rangle = \langle p_{n+1} - \tilde{p}_{n+1}, p \rangle = \langle p, p \rangle,$$

and we deduce $p \equiv 0$.

Finally, in order to prove that each $p \in \mathbb{P}_{n+1}[x]$ is a linear combination of $p_0, \dots, p_{n+1}$, we note that we can always write it in the form $p = cp_{n+1} + q$, where c is the coefficient of x^{n+1} in p and where $q \in \mathbb{P}_n[x]$. The theorem follows by induction. $\square$

Well known examples of orthogonal polynomials include

Legendre polynomials P_n : $[a, b] = [-1, 1]$, $w(x) \equiv 1$;

Chebyshev polynomials T_n : $[a, b] = [-1, 1]$, $w(x) = (1 - x^2)^{-\frac{1}{2}}$;

Laguerre polynomials L_n : $[a, b] = [0, \infty)$, $w(x) = e^{-x}$;

Hermite polynomials H_n : $(a, b) = (-\infty, \infty)$, $w(x) = e^{-x^2}$.